Chapter 7 Homework Assignment

Vectors and Vector Arithmetic

Part A – Python Skills Reinforcement

1. Create vectors a=[2,4,6] and b=[1,3,5]. Print both vectors.

2. Compute vector addition a+b.

3. Compute vector subtraction a–b.

4. Compute element-wise multiplication a*b.

5. Compute element-wise division a/b.

6. Compute the dot product of a and b.

7. Multiply vector a by the scalar 0.5.

8. Create a third vector c=[10,20,30]. Compute a+b+c.

9. Without using NumPy arithmetic operators, manually compute the dot product using a loop.

10. Write a short paragraph explaining the difference between element-wise multiplication and the dot product.

Part B – Practical Problem

A school wants to rank students based on three categories: Homework Score, Quiz Score, Project Score. Represent each student's scores as a vector. Student A = [92, 85, 88] Student B = [80, 95, 90] Student C = [98, 70, 95] The school decides the importance weights are: w = [0.20, 0.30, 0.50] Tasks: 1. Store the three students and the weight vector in NumPy. 2. Use the dot product to compute a weighted score for each student. 3. Rank the students from highest to lowest score. 4. Explain why the dot product is a natural mathematical tool for combining measurements and weights. 5. Suppose the project score becomes twice as important as before. Create a new weight vector and recompute the rankings. 6. Write a short paragraph describing another real-world problem that could be represented with vectors and solved using a dot product.

Challenge Question

Many machine-learning models compute predictions using weighted sums. Explain how the student-ranking problem resembles what a machine-learning model does when it combines features with learned weights.